

COMPLETE CONFIGURATION GUIDE

C-DATA GPON OLT

A COMPLETE STEP-BY-STEP CONFIGURATION GUIDE

From Initial Setup to Service Deployment



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This guide explains the complete configuration process of the C-DATA FD1616S-B1 OLT from its initial state in clear and simple English. It covers Management IP, VLAN, Routing, Uplink, DBA and Traffic Profiles, ONU Registration, and Service Port configuration, together with the required commands and explanations.

This guide is intended primarily for Network Engineers, NOC Engineers, ISP Technicians, and students. The IP addresses and VLANs used here are examples only; always use information approved by your organization in a production network.

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Complete Configuration Guide for the C-DATA OLT: From Initial Setup to Service Deployment

The following sections provide the commands required to configure a new C-DATA FD1616S-B1 OLT and explain the function of each command.

The following example IP addresses are used:

Local OLT IP: 192.168.1.100/24

Laptop IP: 192.168.1.10/24

Production Management VLAN: 112

New OLT IP: 10.49.12.222/30

New Gateway: 10.49.12.221

Retail uplink: XGE 0/0/1

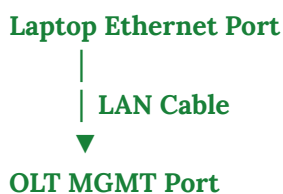
Test customer VLAN: 501

Test GPON port: GPON 1

Caution: The IP addresses 10.49.12.222/30 and 10.49.12.221 are used for demonstration purposes only. Use the actual IP addresses assigned to your network.

Step 1: Connect the Laptop to the OLT

Physical connection:



Configure the following on the laptop:

IP address: 192.168.1.10

Subnet mask: 255.255.255.0

Default gateway: N/A

DNS: N/A

Test from CMD: ping 192.168.1.100

What does it do? It checks whether the OLT at 192.168.1.100 is reachable from the laptop.

If you receive a reply, connect through SSH: `ssh username@192.168.1.100`

Alternatively, open a browser: `http://192.168.1.100`

Step 2: Configuration mode

After logging in to the OLT:

```
# enable  
# configure terminal
```

On some firmware versions:

```
# enable  
# config
```

Command	Function
enable	Enters administrator/privileged mode
configure terminal	Enters configuration mode
config	Short form of the configuration-mode command on some firmware versions

Step 3: OLT hostname

Command:

```
(config)# sysname Ambarkhana_GPON-01
```

Command Explanation:

Sets the OLT hostname. This name appears in monitoring systems, SSH sessions, and configuration backups.

Step 4: Local MGMT IP

Command:

```
Ambarkhana_GPON-01(config)# interface mgmt  
Ambarkhana_GPON-01(config-interface-mgmt)# ip address 192.168.1.100 24  
Ambarkhana_GPON-01(config-interface-mgmt)# exit
```

Command Explanation:

Command	Function
interface mgmt	Enters the dedicated management-port configuration mode
ip address 192.168.1.100 24	Assigns 192.168.1.100/24 to the MGMT port
exit	Exits interface configuration mode

Here, /24 = 255.255.255.0

The laptop and OLT are on the same network:

Laptop: 192.168.1.10/24
OLT: 192.168.1.100/24

10.49.12.222/30 and 10.49.12.221 belong to the same 10.49.12.220/30 network.

Step 5: Create a New Administrator

Command:

```
Ambarkhana_GPON-01(config)# user name noc password ***** privilege-level 15
```

Command Explanation:

Parameter	Meaning
user name noc	Creates a user named noc
password	Sets the login password
privilege-level 15	Assigns the highest administrator privilege level

Step 6: Management services

Command:

```
Ambarkhana_GPON-01(config)# service ssh enable  
Ambarkhana_GPON-01(config)# service https enable  
Ambarkhana_GPON-01(config)# service snmp enable  
Ambarkhana_GPON-01(config)# service http enable
```

Command Explanation:

Command	Function
service ssh enable	Enables secure remote CLI access
service https enable	Enables secure web management
service snmp enable	Enables NMS monitoring
service telnet enable	Enables remote CLI access to the OLT over the network
service http enable	Enables standard web management

Use SSH/HTTPS in a production network. Do not enable Telnet unless it is required.

Step 7: Create VLANs

Complete VLAN list for the OLT:

vlan 1,44,86,110,112,200,221,257,410-411,420
vlan 431,480,501-503,505-508,785-787,934,1120

Command:

Ambarkhana_GPON-01(config)# vlan 1121-1124

Create vlan successfully:

1121-1124

Step 8: Production management IP

Assume the NOC team has assigned:

Management VLAN: 112

OLT IP: 10.49.12.222/30

Gateway: 10.49.12.221

Command:

Ambarkhana_GPON-01(config)# interface vlanif 112

Ambarkhana_GPON-01(config-vlanif-112)# ip address 10.49.12.222 30

Ambarkhana_GPON-01(config-vlanif-112)# exit

Command Explanation:

Command	Function
interface vlanif 112	Creates or opens the Layer 3 interface for VLAN 112

ip address 10.49.12.222 30	Assigns the OLT production IP address to VLAN 112
exit	Exits VLAN interface configuration mode

Here, /30 = 255.255.255.252

Example subnet:

IP	Function
10.49.12.220	Network address
10.49.12.221	Router/Gateway
10.49.12.222	New OLT
10.49.12.223	Broadcast

Step 9: Default route

This command gives the OLT a routing instruction:

ip route 0.0.0.0/0 10.49.12.221

It means: If no more specific route matches, send the packet to the router at 10.49.12.221. Here, 0.0.0.0/0 represents all other destinations, and 10.49.12.221 is the router IP address.

Assume the OLT IP address is 10.49.12.222/30:

Destination	OLT Action
10.49.12.221	Same subnet: the OLT reaches the router directly; no default route is needed.
NMS IP on another network	The OLT sends the packet to the .221 router first. This requires a default route.

The OLT (.222) and router (.221) are on the same subnet, so the OLT can ping the router directly. The NMS is on a remote network, so the OLT must be instructed to use the .221 router to reach it. This instruction is the default route: **ip route 0.0.0.0/0 10.49.12.221**

A default route is not required to reach the directly connected router; it is required to reach networks beyond that router. In short, the default route tells the OLT which router to use for other networks.

Step 10: Retail uplink configure

Command:

```
Ambarkhana_GPON-01(config)# interface xge 0/0
Ambarkhana_GPON-01(config-xge-0/0)# vlan mode 1 trunk
Ambarkhana_GPON-01(config-xge-0/0)# vlan trunk 1 44,86,110,112,200,221,257,410-411,420,480

Ambarkhana_GPON-01(config-xge-0/0)# port 1 trust-mode both
Ambarkhana_GPON-01(config-xge-0/0)# exit
Ambarkhana_GPON-01(config)# save
```

Command Explanation:

Command	Function
<code>interface xge 0/0</code>	Enters the 10G uplink interface group
<code>vlan mode 1 trunk</code>	Configures XGE port 1 as a trunk
<code>vlan trunk 1 ...</code>	Defines which VLANs are allowed through port 1
<code>exit</code>	Exits XGE interface configuration mode

Management VLAN 112 must be included in the trunk list.

Step 11: Corporate uplink configure

corporate uplink: GE 0/0/3, If a corporate uplink is not required, you do not need to configure it

Command:

```
Ambarkhana_GPON-01(config)# interface ge 0/0
Ambarkhana_GPON-01(config-xge-0/0)# vlan mode 3 trunk
Ambarkhana_GPON-01(config-xge-0/0)# vlan trunk 3 44,86,110,112,200,221,257,410-411,420,431
Ambarkhana_GPON-01(config-xge-0/0)# port 3 trust-mode both
Ambarkhana_GPON-01(config-xge-0/0)# exit
Ambarkhana_GPON-01(config)# save
```

Command Explanation:

Command	Function
<code>interface ge 0/0</code>	Enters the Gigabit Ethernet uplink group
<code>port-desc 3</code>	Sets the description for GE port 3
<code>vlan mode 3 trunk</code>	Configures GE port 3 as a trunk
<code>vlan trunk 3 ...</code>	Defines the VLANs allowed through GE port 3
<code>port 3 trust-mode both</code>	Trusts QoS markings on port 3

Step 12: Configure the Other End of the Uplink

The same VLANs must be allowed on the switch/router port connected to the OLT XGE port:

OLT XGE 0/0/1:

VLAN 112,501,502,503,506,507,508 allowed

Switch uplink port:

VLAN 112,501,502,503,506,507,508 allowed

Traffic will not pass if the VLAN is allowed on only one side of the link.

Step 13: Gateway test

After connecting the uplink, run the following command from the OLT:

ping 10.49.12.221

Function: Checks whether the production gateway is reachable from the OLT. If the ping fails, verify the following:

- Whether the uplink is up
- Whether VLAN 112 is allowed on both ends
- Whether the OLT IP and gateway are in the same subnet
- Whether the switch port is configured as a trunk
- Whether the gateway interface is up

Step 14: DBA Profile

The DBA profile controls GPON upstream bandwidth allocation.

Command:

30 Mbps DBA:

```
Ambarkhana_GPON-01(config)# dba-profile profile-id 30 profile-name dba-profile_30
Ambarkhana_GPON-01(config-dba-profile-30)# type4 max 30720
Ambarkhana_GPON-01(config-dba-profile-30)# commit
Ambarkhana_GPON-01(config-dba-profile-30)# exit
```

60 Mbps DBA:

```
Ambarkhana_GPON-01(config)# dba-profile profile-id 60 profile-name dba-profile_60
Ambarkhana_GPON-01(config-dba-profile-60)# type4 max 61440
Ambarkhana_GPON-01(config-dba-profile-60)# commit
Ambarkhana_GPON-01(config-dba-profile-60)# exit
```

Maximum 1 Gbps DBA:

```
Ambarkhana_GPON-01(config)# dba-profile profile-id 1 profile-name dba-profile_1
Ambarkhana_GPON-01(config-dba-profile-1)# type4 max 1024000
Ambarkhana_GPON-01(config-dba-profile-1)# commit
Ambarkhana_GPON-01(config-dba-profile-1)# exit
```

Command Explanation:

Command	Function
dba-profile profile-id 30	Creates DBA profile ID 30
profile-name dba-profile_30	Sets the profile name
type4	Best-effort dynamic bandwidth allocation
max 30720	Sets the maximum rate to approximately 30 Mbps
commit	Applies the profile changes
exit	Exits the profile configuration

Conversion:

30 Mbps × 1024 = 30720 Kbps
60 Mbps × 1024 = 61440 Kbps
100 Mbps × 1024 = 102400 Kbps

Step 15: Traffic Profile

Command:

```
Ambarkhana_GPON-01(config)# traffic-profile profile-id 30 profile-name "30MB" cir 30720 pir 30720
cbs 2000 pbs 2000

Ambarkhana_GPON-01(config)# traffic-profile profile-id 60 profile-name "60MB" cir 61440 pir 61440
cbs 2000 pbs 2000

Ambarkhana_GPON-01(config)# traffic-profile profile-id 100 profile-name "100MB" cir 102400 pir
102400 cbs 2000 pbs 2000
```

Command Explanation:

Parameter	Function
profile-id	Unique profile ID
profile-name	Profile name
cir	Committed Information Rate
pir	Peak Information Rate
cbs	Committed Burst Size
pbs	Peak Burst Size

Example:

cir 30720 pir 30720 means that both the committed and maximum rates are approximately 30 Mbps.

A 100 Mbps client is affected by three separate rules:

MikroTik Simple Queue = the client's package is 100 Mbps
ONU ETH 1 Traffic Profile 255 = a maximum of 500 Mbps in both directions
OLT DBA Profile 30 = a maximum of approximately 30 Mbps for GPON upstream traffic

Configuration:

```
Ambarkhana_GPON-01(config)# interface gpon 0/0  
Ambarkhana_GPON-01(config-gpon-0/0)# ont port car 3 2 eth 1 inbound 255 outbound 255
```

This means that Profile ID 255 is applied to both upstream and downstream traffic on Ethernet port 1 of ONU 2 on GPON port 3. According to the profile list, ID 255 represents a 500 Mbps profile, not 255 Mbps.

Therefore, Traffic Profile 500 does not reduce the client's 100 Mbps package; the 100 Mbps MikroTik limit remains effective. However, if DBA 30 is active on that ONU, upstream traffic may be limited to approximately 30 Mbps. The same conclusion cannot be applied to downstream traffic based on DBA 30 alone.

Remember: MikroTik 100 Mbps, Traffic Profile 500 Mbps, and DBA upstream 30 Mbps means that 30 Mbps is the smallest limiting value in the upstream path.

Step 16: VLAN 504 line profile

Command:

```
Ambarkhana_GPON-01(config)# ont-line-profile gpon profile-id 504 profile-name lineprofile_504  
Ambarkhana_GPON-01(config-ont-lineprofile-504)# tcont 1 dba-profile-id 30  
Ambarkhana_GPON-01(config-ont-lineprofile-504)# mapping-mode vlan  
Ambarkhana_GPON-01(config-ont-lineprofile-504)# gem add 1 tcont 1  
Ambarkhana_GPON-01(config-ont-lineprofile-504)# gem mapping 1 1 vlan 504  
Ambarkhana_GPON-01(config-ont-lineprofile-504)# gemport 1 tag-action vlan-transparent  
Ambarkhana_GPON-01(config-ont-lineprofile-504)# commit  
Ambarkhana_GPON-01(config-ont-lineprofile-504)#exit
```

Command Explanation:

Command	Function
ont-line-profile gpon...	Creates a GPON ONU line profile
tcont 1 dba-profile-id 30	Associates the 30 Mbps DBA profile with T-CONT 1
mapping-mode vlan	Maps traffic to the GEM port according to VLAN

<code>gem add 1 tcont 1</code>	Associates GEM port 1 with T-CONT 1
<code>gem mapping 1 1 vlan 504</code>	Maps VLAN 504 traffic to GEM port 1
<code>tag-action vlan-transparent</code>	Passes the VLAN tag without modification
<code>commit</code>	Saves and applies the profile

Step 17: Configure Manual PON Authentication

Command:

```
Ambarkhana_GPON-01(config)# interface gpon 0/0
Ambarkhana_GPON-01(config-gpon-0/0)# ont authmode all manual
Ambarkhana_GPON-01(config-gpon-0/0)# exit
```

Command Explanation:

Instead of allowing automatic ONU registration, each ONU must be registered manually by an administrator on every GPON port.

Example:

ont authmode 5 manual means that ONU registration on GPON port 5 will be manual.

Step 18: Find an Unregistered ONU

Command:

```
Ambarkhana_GPON-01(config-gpon-0/0)# show ont autofind all
```

Command Explanation:

Displays the serial number and PON port of ONUs detected by the OLT but not yet registered.

Step 19: ONU Configuration

Command:

```
Ambarkhana_GPON-01(config-gpon-0/0)# ont add 2 42 sn-auth CDTCAFA95745ont-lineprofile-id
501 ont-srv-profile-id 4
Ambarkhana_GPON-01(config-gpon-0/0)# ont port native-vlan 2 42 eth 1 vlan 501 priority 0
Ambarkhana_GPON-01(config-gpon-0/0)# ont port car 2 42 eth 1 inbound 255 outbound 255
Ambarkhana_GPON-01(config-gpon-0/0)# exit
```

```
Ambarkhana_GPON-01(config)# service-port 333 vlan 503 gpon 0/0 port 2 ont 48 gemport 1  
multi-service user-vlan 501 tag-action transparent
```

Step 20: Create a Service Port

Command:

```
Ambarkhana_GPON-01(config-gpon-0/0)# service-port autoindex vlan 501 gpon 0/0 port 2 ont 42  
gemport 1 multi-service user-vlan 501 tag-action transparent
```

Command Explanation:

Command	Function
service-port autoindex	Unique service-port ID
vlan 501	Network-side VLAN
gpon 0/0 port 2	GPON port 2
ont 42	ONU ID 42
gemport 1	GEM port 1
user-vlan 501	ONU/customer-side VLAN
tag-action transparent	Keeps the VLAN tag unchanged

Traffic Path:

```
ONU 1  
↓ VLAN 501  
GPON port 1  
↓ GEM port 1  
Service-port 1  
↓ VLAN 501  
XGE uplink
```

Every new service-port ID must be unique. The same ID cannot be used more than once.

Step 21: Check ONU Status

All ONU Information:

Command:

```
Ambarkhana_GPON-01(config-gpon-0/0)# show ont info all
```

Command Explanation:

Displays the online/offline and configuration status of all registered ONUs.

ONU Laser/dB level Check:

Command:

```
Ambarkhana_GPON-01(config-gpon-0/0)# show ont optical-info all  
Ambarkhana_GPON-01(config-gpon-0/0)# show ont optical-info 2 all (specific pon optical power  
check)
```

Command Explanation:

Displays the ONU receive/transmit optical information.

Step 22: Check MAC Addresses

Command:

```
Ambarkhana_GPON-01(config)# show mac-address vlan 501
```

Command Explanation:

Checks whether the customer/router MAC address has been learned in VLAN 501.

If a MAC address is found, it confirms the following path:

ONU → GPON → Service-port → VLAN → Uplink

The Layer 2 traffic path is working.

Step 23: Check Routing

```
Ambarkhana_GPON-01(config)# show ip route
```

Expected default route:

```
0.0.0.0/0 via 10.49.12.221
```

Remote test:

```
ping 10.49.12.221
```

If an NMS IP address is available:

```
ping NMS_IP
```

In simple terms:

An NMS is a computer/server used to monitor whether the OLT is online or offline.

Example: OLT -> Gateway -> NMS

First, run ping 10.49.12.221 to check connectivity from the OLT to the gateway.

Then, if the actual NMS IP address is known:

ping NMS_IP_ADDRESS

This checks connectivity from the OLT to the monitoring server.

For example, if the NMS IP is 10.10.10.50, run: ping 10.10.10.50

NMS_IP is a placeholder, not an actual IP address. Replace it with the real IP address of your LibreNMS, Zabbix, or The Dude server.

Step 24: Configure Timezone, DNS, and NTP

Command:

```
Ambarkhana_GPON-01(config)# timezone gmt+ 06:00
Ambarkhana_GPON-01(config)# dns server 123.200.0.254
Ambarkhana_GPON-01(config)# dns server-bak 8.8.8.8
Ambarkhana_GPON-01(config)# ntp-service refclock-master server 203.76.96.95
Ambarkhana_GPON-01(config)# ntp-service refclock-master server-bak 123.200.0.252
Ambarkhana_GPON-01(config)# ntp-service client enable
Ambarkhana_GPON-01(config)# save
```

Command Explanation:

Command	Function
timezone gmt+ 06:00	Sets the Bangladesh timezone
dns server	Sets the primary DNS server
dns server-bak	Sets the backup DNS server
ntp...server	Sets the primary time server
server-bak	Sets the backup time server
client enable	Enables NTP synchronization

Step 25: Verify and Save the Configuration

Command:

```
Ambarkhana_GPON-01(config)# show current-config  
Ambarkhana_GPON-01(config)# save
```

When prompted for confirmation, enter: y

Command Explanation:

The save command stores the running configuration as the startup configuration. If you do not save it, the configuration may be lost after a reboot.